Set A

BRAC UNIVERSITY

Department of Computer Science and Engineering

Examination: Semester Midterm — Semester: Summer 2025

Duration: 1 Hour 30 Minutes — Full Marks: 40

CSE 422: Artificial Intelligence

Answer all the following questions. **(Keep your answers precise and to the point)**

Figures in the right margin indicate marks.

Name:	ID:	Section:

1. CO1 A city is preparing to deploy six different AI surveillance systems across three urban zones. These include facial recognition, license plate readers, sound detection, drone tracking, crowd estimation, and weapon detection. While each system serves a specific public safety role, deploying certain combinations in the same zone raises ethical concerns—such as privacy violations, over-surveillance, and potential discrimination.

City regulators have identified specific system pairs that must not be deployed together. Combining facial recognition with license plate readers or sound detection can lead to invasive tracking. Crowd estimation should not run alongside facial recognition or sound detection to avoid targeting certain populations. Other conflicts include license plate readers with drone tracking, sound detection with either drone tracking or crowd estimation, and drone tracking with weapon detection.

To ensure ethical compliance, each AI system must be assigned to one of three fixed zones in a way that avoids conflicts. The city has chosen to use a Genetic Algorithm to find an optimal deployment strategy that respects all these constraints.

a. How would you **encode** a chromosome to represent an assignment of the six AI systems to the three zones? 2.5

b. **Design** a fitness function that: 2.5
Penalizes each conflict (i.e., when two incompatible systems are assigned to the same zone), and
Encourages conflict-free assignments using the available zones.

c. **Provide** a simple example of a chromosome and compute its fitness using your proposed function. 2.5

d. **Discuss** the issue that may arise if the mutation rate is too high or too low. 2.5

2. CO1

S			R	G
	#			
	#	R	R	R
	#			
		R		

Inside a factory, autonomous mobile robots (AMRs) are responsible for transporting raw materials and finished components between different production stations. Imagine the factory floor as a **5 by 5 grid**, where jumping from one cell to another cost 1 unit but hazardous zones are marked as R which cost 10 units and the blocked area are marked as #.
S(1,1) represents the starting position of the AMR and **G(1,5)** represents the destination.

a. Is it possible to **define** a heuristic that is consistent yet not informative/useful to the informed search algorithms? Justify your answer with example 3

b. How effective do you think Euclidean Distance will be as a heuristic function for the following Grid? **Support** your argument with reason 3

c. **Define** a suitable heuristic on your own and simulate graph A* search from **start state S** to **Goal Node G**. 4

3. CO1 **State** whether the following statements are true/false with brief explanations. **The explanation can be very brief but no marks will be given without explanation**

a. Simulated annealing's probability of denying worse solutions increases as the search progresses. 2.5

b. Simulated annealing guarantees better performance than hill climbing in all types of search spaces. 2.5

c. Simulated annealing becomes equivalent to hill climbing when the initial temperature is very high. 2.5

d. When optimizing a function with many local maxima of similar values, random restarts combined with hill climbing will never increase the likelihood of finding the global optimum. 2.5

4. CO1

O	O	X
	X	
O	X	

Suppose you are to design an agent **(circle)** that knows how to play a game of tic tac toe against the user **(cross)**. The above figure shows the current state of the game. Now the agent has to decide which move to make by employing alpha beta pruning. Therefore answer the following questions based on this scenario

a. **Determine** which move would the agent prefer from the given state of the game. The final utilities will be **[(100 - Depth) * 1]** for a win, **[(100 - Depth) * -1]** for a loss and 0 for a draw. You must showcase the values of the nodes, **α** and **β**. 7

*[Hint: Assuming the depth of the root node is 1, if a leaf node is found at the next level (depth 2) where the agent wins, then the utility for that leaf node from the agent's perspective would be (100 - 2) *1 = 98. For a losing state it would have been (100 - 2) * -1 = -98]*

b. "Any state of a tic tac toe game tree can be used to show that it is a zero sum environment". **Judge** the validity of this statement with mathematical calculations. 3

[Hint : You have to prove your opinion using any two states of your choice from the game tree derived from part a]